Department of Materials Science and Engineering

B.Tech. Major Project Part-1 / Part-II (MLD 411/412)

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Project Title: Techno-Economic Assessment of Retrofit Pathways for Low-Carbon Coal-Based Rotary-Kiln DRI

Objectives:

- Develop a first-principles computational digital twin and techno-economic simulation framework of an industrial coal-based rotary-kiln Direct Reduced Iron (sponge iron) plant.
- Simultaneously resolve multi-component mass and enthalpy balances across the kiln bed, freeboard, and after-burner chamber boundaries without empirical fitting factors.
- Evaluate technical, carbon reduction (targeting 20%, 40%, and 60% CO_2 abatement), and economic feasibility (Rs./t DRI, payback period, and break-even hydrogen price) of retrofits: waste heat recovery (WHR), biomass/biocarbon substitution, and hydrogen injection.

Summary of work done:

For the mid-term milestone of this project, the core computational architecture and baseline thermodynamic digital twin of an industrial coal-based rotary-kiln Direct Reduced Iron (sponge iron) plant have been successfully developed and computationally verified. Operating entirely on first-principles conservation laws rather than empirical fitting constants, the model simultaneously resolves coupled non-linear mass and enthalpy balances across three interconnected thermodynamic control volumes which is the counter-current reactive solid bed, the gaseous freeboard, and the after-burner chamber. The model enforces atom-level conservation across seven distinct elemental streams, encompassing iron, carbon, hydrogen, nitrogen, sulfur, oxygen, and inert coal minerals, while mechanistically accounting for the progressive reduction of iron oxides, endothermic flux calcination, volatile matter post-combustion, and the temperature-dependent Boudouard gasification equilibrium. By treating specific coal consumption, process air flow rates, and waste-gas generation as solved system variables derived directly from physical conservation constraints, the baseline formulation achieves an exceptionally tight mass balance closure with an overall residual of approximately one percent and a thermal ledger discrepancy constrained to well under one-hundredth of a percent. In parallel, a comprehensive inventory of over ninety verified thermochemical and operational parameters has been established, alongside initial mathematical formulations of the primary retrofit modules, waste heat recovery, biomass co-firing, and hydrogen injection, integrated into an interactive computational dashboard for preliminary techno-economic and carbon emissions accounting.

Future plan:

Building upon the validated baseline framework established during the initial phase, the subsequent stage of the project will focus on comprehensive scenario execution, parametric sensitivity analysis, and multi-objective optimization to establish actionable industrial decarbonization pathways. Immediate efforts will involve deploying automated parametric sweeps across critical techno-economic variables, most notably fluctuating coal and electricity tariffs, biocarbon supply costs, and hydrogen delivery pricing, to quantify their direct influence on production costs per tonne of product and determine empirical break-even thresholds for low-carbon fuels. Furthermore, the analysis will be systematically scaled across commercial plant capacities ranging from decentralized one hundred tonne-per-day units typical of small and medium enterprises to large industrial kilns, thereby evaluating how economy of scale dictates the financial viability of heat recovery versus fuel substitution. The core analytical milestone will entail formulating a computational optimization routine that determines the mathematically least-cost combination of retrofit

measures capable of achieving predefined carbon abatement targets of twenty, forty, and sixty percent without degrading product metallization or kiln throughput. Finally, these optimization outcomes will be synthesized into a continuous Marginal Abatement Cost Curve and an actionable industrial retrofit roadmap, alongside targeted thermodynamic refinements addressing auxiliary electrical parasitics and high-ash coal behavior to deliver an exhaustive, industry-ready decision-support tool.

Comments on Performance:

Satisfactory [✓] Average [ ] Unsatisfactory [ ]

Name of Supervisor: Prof. Suryanarayana Vikrant Karra **Signature of Supervisor:** S. V. Vikrant

Date: _____